Research Project Report

科研项目报告

Perturbation-informed mapping reveals *RAC1*-responsive tumor-state organization in lung adenocarcinoma

扰动信息驱动的状态映射揭示肺腺癌中 *RAC1* 响应性肿瘤状态组织

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Highlights

- scTenifoldKnk-based virtual knockout was used to identify *RAC1*-responsive regulatory rewiring genes in LUAD tumor cells.
- *RAC1*-responsive genes reconstructed a perturbation-informed latent tumor-state manifold from WT single-cell tumor cells.
- The latent manifold revealed hybrid tumor states and continuous functional program gradients.
- *RAC1*-responsive genes retained prognostic relevance in the TCGA-LUAD cohort.
- This project demonstrates a computational perturbation framework linking GRN rewiring to tumor-state mapping.
- A supplementary Python-based graph representation analysis further supported network-role shifts of *RAC1*-responsive genes using topology metrics, adjacency-vector PCA embedding, and aligned node2vec embedding.

Abstract

Background

Lung adenocarcinoma (LUAD) exhibits substantial tumor-state heterogeneity and cellular plasticity. Distinct tumor-cell states are often characterized by different combinations of functional programs, yet the regulatory network architecture underlying these latent states remains incompletely understood. Conventional differential expression analysis can identify genes with altered expression levels, but it does not fully capture perturbation-associated changes in regulatory network topology or gene network roles.

RAC1, a member of the Rho family GTPases, participates in cytoskeletal remodeling, migration, EMT, invasion, and therapy resistance. These functions make *RAC1* a biologically meaningful perturbation anchor for studying tumor-state plasticity in LUAD. However, whether *RAC1*-responsive regulatory rewiring can help organize or explain LUAD tumor-cell state architecture at the single-cell level remains unclear.

Methods

Based on LUAD single-cell RNA-seq dataset GSE131907 as the primary dataset, this project developed a perturbation-informed computational framework integrated scTenifoldKnk-based virtual knockout, gene regulatory network (GRN) rewiring analysis, perturbation-responsive gene-based latent tumor-state manifold reconstruction, UCell functional program scoring, and TCGA-LUAD prognostic modeling. *RAC1*-responsive genes derived from virtual knockout analysis were used as graph-informed features to reconstruct a latent tumor-state manifold in unperturbed tumor cells. Functional program landscapes, hybrid state annotations, and continuous program gradients were then analyzed within this perturbation-informed state space.

Results

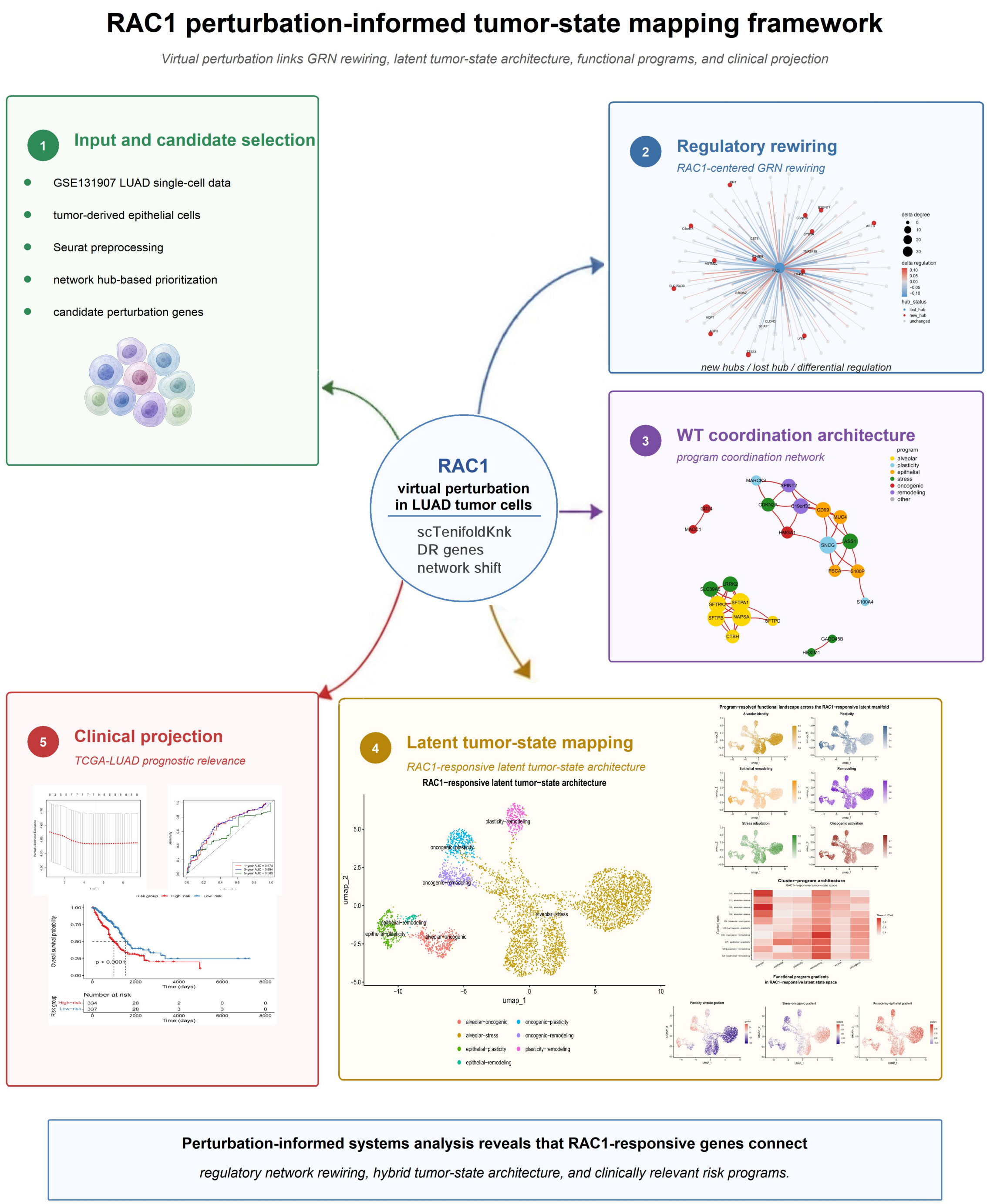
Among virtual perturbation targets, *RAC1* showed the strongest network-disruptive response, with the highest mean network distance (3.44×10^{-5}), maximum distance (0.001713), and number of significant differentially rewired genes ($n = 13$). *RAC1*-specific GRN rewiring analysis produced a network containing 141 genes and 140 rewired regulatory edges, revealing hub-status reorganization after virtual perturbation.

Using 99 *RAC1*-responsive genes, WT tumor cells were reconstructed into a perturbation-informed latent manifold containing 10 clusters, which were further annotated into 7 hybrid functional states. UCell-based program scoring showed that alveolar identity, plasticity, epithelial, stress, oncogenic, and remodeling programs were spatially organized across the latent manifold. Functional gradient analysis further suggested that these tumor states were not fully discrete clusters, but were organized along continuous functional axes.

Finally, an eight-gene *RAC1*-responsive prognostic signature stratified TCGA-LUAD patients by survival risk, and the risk score remained an independent prognostic factor in multivariate Cox analysis ($HR = 2.11$, 95% $CI = 1.74–2.57$, $p = 8.03 \times 10^{-14}$).

Conclusion

This study establishes a perturbation-informed latent state mapping framework that links *RAC1* virtual perturbation, GRN rewiring, and single-cell tumor-state organization. The results suggest that *RAC1*-responsive genes may contribute to transcriptional coordination, hybrid functional state organization, and continuous program gradients in LUAD tumor cells. This project provides a computational systems biology strategy for moving from single-cell expression profiling toward perturbation-informed tumor-state mapping.



Graphical Overview. ***RAC1*** **perturbation-informed tumor-state mapping framework.** The project starts from LUAD single-cell tumor epithelial cells and network hub-based candidate perturbation selection. *RAC1* virtual perturbation was then performed using scTenifoldKnk to identify perturbation-responsive genes and regulatory network rewiring. *RAC1*-responsive genes were further used to construct a WT coordination network and a perturbation-informed latent tumor-state manifold. Functional program scoring revealed hybrid tumor-state architecture and continuous program gradients, while TCGA-LUAD analysis evaluated the clinical relevance of *RAC1*-responsive programs.

1. Introduction

1.1 Tumor-state heterogeneity and plasticity

Tumor cells do not exist as fixed or uniform entities. Instead, they can occupy diverse and plastic functional states shaped by different combinations of transcriptional programs. Recent single-cell transcriptomic studies have revealed substantial tumor-state heterogeneity within LUAD, including alveolar-like, stress-like, epithelial-like, and plasticity-associated states. These states not only reflect transcriptional diversity, but may also be associated with invasion, metastasis, therapeutic tolerance, and disease progression[1].

Increasing evidence suggests that tumor-state space is not always organized as a set of fully discrete cell populations. Rather, tumor cells may display hybrid state identities shaped by combinatorial activities of multiple functional programs and continuous transcriptional gradients[2, 3]. Therefore, understanding LUAD tumor-state organization at a systems level requires moving beyond individual marker genes or cluster labels. It requires a framework that can capture how regulatory and functional programs are coordinated within latent tumor-state space.

1.2 Perturbation biology and regulatory network rewiring

In single-cell analysis, comparing gene expression levels alone is often insufficient to explain the regulatory basis of cell-state differences. Differences between tumor states may reflect not only marker-gene expression, but also changes in regulatory interactions, network topology, and network roles of genes. Compared with conventional differential expression analysis, regulatory network rewiring provides a complementary perspective for characterizing how gene regulatory interactions and transcriptional coordination patterns change under perturbation.

Recent computational perturbation approaches have begun to integrate single-cell expression profiles, gene regulatory network inference, and perturbation modeling to study cell-state regulation. For example, scTenifoldKnk performs in silico virtual knockout analysis to identify genes with altered network roles after perturbation[4]. CellOracle integrates GRN inference with in silico gene perturbation to predict how transcription factor perturbations may affect cell identity and state behavior[5]. GEARS uses geometric deep learning to model transcriptional responses to single-gene and combinatorial perturbations[6]. Together, these methods reflect a broader shift from detecting expression changes toward modeling perturbation responses and their regulatory network basis.

This perspective provides a useful computational entry point for studying tumor-state organization. If perturbation-induced GRN rewiring can be translated into informative features for single-cell state mapping, it may help reveal how network-level perturbation responses are associated with latent tumor-cell state architecture. In this project, *RAC1* virtual perturbation was used as a biologically motivated anchor to explore whether *RAC1*-responsive regulatory rewiring genes can define a perturbation-informed tumor-state space in LUAD.

1.3 *RAC1* in lung adenocarcinoma

RAC1 is a member of the Rho family small GTPases and plays important roles in cytoskeletal remodeling, cell migration, adhesion, EMT, invasion, and therapy resistance. *RAC1* has been implicated in cancer cell motility, invasion, metastasis, and adaptive responses to stress[7, 8]. In non-small cell lung cancer, *RAC1* overexpression has been associated with EMT features and poor prognosis, suggesting a potential role in tumor-cell plasticity during lung cancer progression[9]. Recent single-cell transcriptomic analysis of LUAD brain metastasis further identified *RAC1* as a potential key factor promoting brain metastasis, supporting its relevance to tumor-cell state remodeling in LUAD[10].

Based on these findings, this project used *RAC1* as a perturbation anchor to investigate whether *RAC1*-responsive regulatory network rewiring can inform the functional organization of LUAD tumor-cell state space.

1.4 Study objective

The core question of this project is whether *RAC1* virtual perturbation-derived regulatory network rewiring can be used to interpret latent tumor-state organization in LUAD. Rather than focusing on *RAC1* expression or prognostic association alone, this study treats *RAC1* as a computational perturbation anchor and asks whether *RAC1*-responsive genes can function as perturbation-informed features for reconstructing a latent tumor-state manifold in WT tumor cells.

Specifically, *RAC1* virtual knockout was used to identify *RAC1*-responsive genes. These genes were then used to reconstruct a latent tumor-state manifold, annotate hybrid functional states, analyze continuous functional gradients, and assess clinical relevance in the TCGA-LUAD cohort. Through this framework, the project links GRN rewiring and single-cell state mapping to explore how *RAC1*-associated perturbation responses may contribute to LUAD tumor-state organization.

2. Methods

2.1 Single-cell RNA-seq preprocessing and candidate gene selection

The LUAD single-cell RNA-seq dataset GSE131907 was used as the primary dataset for this study[1]. The original UMI count matrix and cell annotation file were processed according to the sample origin and refined cell-type annotation. Tumor tissue-derived epithelial cells were selected using Sample_Origin = tLung and Cell_type.refined = Epithelial cells.

To reduce computational burden for virtual knockout and network construction, 5,000 tumor epithelial cells were randomly downsampled. Genes detected in at least 25 cells were retained for downstream analysis. A Seurat object was then constructed, followed by log-normalization and identification of highly variable genes (HVGs)[11]. In parallel, the average expression level of each retained gene was calculated across the selected tumor cells, and the top 2,000 highly expressed genes were selected. The intersection between the 2,000 HVGs and the top 2,000 highly expressed genes was used as the initial candidate gene set for virtual perturbation and GRN analysis.

2.2 Network hub-based prioritization

After obtaining the initial candidate gene set, Pearson expression correlations were computed among candidate genes across the selected tumor cells. The absolute correlation coefficients were used as edge weights to construct an undirected weighted gene co-expression network. This network was used as a preliminary network-informed prioritization layer for candidate perturbation target selection.

Within this network, degree centrality and betweenness centrality were calculated for each gene to estimate network connectivity and potential bridging roles. The sum of degree centrality and betweenness centrality was used as a composite network hub score. Genes ranked highly by this score were considered candidate network hubs. This step was not intended as a fully automated target-selection pipeline, but rather as a network-guided prioritization procedure to support perturbation target selection.

Based on network centrality, expression detectability, suitability for virtual knockout comparison, and known tumor-related biological functions, 10 candidate perturbation genes were selected for downstream virtual knockout screening: *RAC1*, *LDHA, AREG, ANXA2, S100A6, LPCAT1, TMSB10, TAGLN2, CTSD*, and *PON2*.

2.3 Multi-gene virtual perturbation screening

For each selected candidate perturbation gene, in silico virtual knockout analysis was performed using scTenifoldKnk[4]. scTenifoldKnk constructs single-cell gene regulatory networks under WT and virtual knockout conditions and identifies genes with perturbation-associated changes in regulatory relationships or network roles. In this study, these genes are referred to as differentially rewired genes (DR genes). The main scTenifoldKnk parameters were set as qc = TRUE, nc_nNet = 8, and nc_nCells = 500.

To compare the network-level impact of different virtual perturbation targets, the top 100 DR genes ranked by perturbation distance were extracted from each virtual knockout result. For each perturbation target, three summary metrics were calculated: mean distance, maximum distance, and the number of significant DR genes with adjusted p value < 0.05. Mean distance and maximum distance were used to estimate the overall network shift and the strongest local perturbation-associated change, respectively, while the number of significant DR genes was used to reflect the breadth of perturbation effects.

The perturbation target showing the strongest overall network response across these metrics was selected for downstream GRN rewiring analysis, latent manifold reconstruction, and prognostic relevance analysis.

2.4 *RAC1*-specific GRN rewiring and coordination network analysis

Based on the multi-gene virtual perturbation screening, *RAC1* was selected as the core perturbation target for downstream analysis. To evaluate the effect of *RAC1* virtual perturbation on inferred regulatory network topology, WT and *RAC1*-vKO tensor regulatory networks were extracted from the *RAC1* virtual knockout result and converted into adjacency matrices. A delta regulation matrix was then calculated by subtracting the WT network from the *RAC1*-vKO network. After symmetrization, this matrix was used to quantify *RAC1*-vKO-associated regulatory network rewiring.

To construct the GRN rewiring network shown in Figure 1A, delta regulation values were thresholded and isolated nodes were removed. Because a small number of extreme *RAC1*-associated delta edges could dominate the network layout, quantile-based clipping was applied to *RAC1*-associated edges for visualization only. In the resulting network, nodes represent genes, and edges represent inferred regulatory relationships that changed between WT and

RAC1-vKO conditions. Node size indicates the absolute degree change, while edge color represents the direction and magnitude of differential regulation. Degree centrality was calculated separately in WT and *RAC1*-vKO networks, and nodes were classified as new hub, lost hub, or unchanged according to degree changes.

To further examine whether *RAC1*-responsive genes already showed functionally interpretable expression coordination in unperturbed WT tumor cells, an expression coordination network was constructed. Top-ranked *RAC1*-responsive genes, excluding *RAC1* itself, were combined with predefined functional program genes, and their expression correlations were calculated across WT tumor cells. This correlation-based coordination network was used to visualize baseline expression relationships among genes associated with alveolar identity, plasticity, epithelial, stress response, oncogenic activation, and remodeling-related programs (Figure 1B). The detailed composition of these functional program gene sets is described in Section 2.6.

2.5 *RAC1*-responsive feature selection and perturbation-informed latent manifold reconstruction

DR genes from *RAC1* virtual knockout were ranked by perturbation distance. Because these DR genes reflect perturbation-associated changes in regulatory relationships or gene network roles rather than conventional expression-level differences, they were referred to as *RAC1*-responsive genes or perturbation-responsive genes in downstream analyses.

To construct a perturbation-informed feature space, the top 100 *RAC1*-responsive genes were selected from the ranked DR gene list. After excluding *RAC1* itself, 99 genes were matched to the WT tumor-cell expression matrix and used for latent tumor-state manifold reconstruction. Based on the expression matrix of these 99 genes in WT tumor cells, a new Seurat object was constructed, followed by normalization, scaling, PCA, and UMAP dimensionality reduction[11, 12]. Nearest-neighbor graph construction and unsupervised clustering were then performed in the same low-dimensional space. The first 10 principal components were used for PCA and UMAP analysis, and the clustering resolution was set to 0.4.

Unlike conventional single-cell embeddings constructed from HVGs, this latent tumor-state manifold was reconstructed from *RAC1*-responsive genes derived from GRN rewiring analysis. Therefore, the purpose of this manifold was not to recapitulate global transcriptomic heterogeneity, but to represent tumor-cell state distributions within a *RAC1*-responsive feature space.

2.6 Functional program scoring and state annotation

To interpret the functional organization of the *RAC1*-responsive latent manifold, six marker-based functional programs were curated based on *RAC1* DR genes, tumor-state-related biological functions, and known marker-gene expression patterns. These programs included alveolar identity, plasticity, epithelial, stress response, oncogenic activation, and remodeling programs.

The gene sets were defined as follows: the alveolar identity program included *SFTPA1, SFTPA2, SFTPB, SFTPD, NAPSA*, and *CTSH;* the plasticity program included *S100A4, MARCKS, KRT17*, and *SNCG;* the epithelial program included *PSCA, S100P, MUC4,* and *CD99;* the stress response program included *GADD45B, DDIT3, HEXIM1, ASS1, SLC39A8, LRRK2,* and

CDKN2A; the oncogenic activation program included *HMGA1, MACC1, PRSS23*, and *CD24;* the remodeling program included *SPINT2* and *C19orf33*.

UCell was then used to score functional program activity in each cell[13]. UCell calculates gene signature scores based on within-cell gene ranking, making it more robust to sequencing depth variation and dropout effects than simple average-expression-based scoring. Because the Seurat object used for latent manifold reconstruction contained only *RAC1*-responsive genes, each functional program gene set was intersected with the manifold feature set before scoring. Therefore, the resulting UCell scores represent relative functional program activity within the *RAC1*-responsive feature space, rather than complete pathway activity at the whole-transcriptome level.

After obtaining cell-level UCell scores, the average program activity of each cluster was calculated and standardized using cluster-level z-scores. Each cluster was initially annotated as a hybrid functional state according to the two programs with the highest z-scores. Since the order of the two high-activity programs does not indicate temporal direction or state transition direction, equivalent labels were merged; for example, stress-alveolar and alveolar-stress were unified as alveolar-stress. The final state annotations were used for latent manifold visualization in Figure 2A and the cluster-program heatmap in Figure 2C.

2.7 Functional program gradient analysis

To further describe relative program enrichment across the *RAC1*-responsive latent manifold, functional program gradients were constructed from UCell program scores. Based on the program landscape and cluster annotations, three biologically interpretable program pairs were selected: plasticity–alveolar, stress–oncogenic, and remodeling–epithelial.

Each gradient was calculated as the difference between two corresponding UCell scores. The plasticity–alveolar gradient was defined as plasticity score minus alveolar score, where positive values indicate relatively higher plasticity activity and negative values indicate relatively higher alveolar identity activity. Similarly, the stress–oncogenic gradient and remodeling–epithelial gradient were used to represent relative enrichment biases between stress versus oncogenic programs and remodeling versus epithelial programs, respectively.

Importantly, no real *RAC1* perturbation single-cell dataset, time-series data, lineage tracing, or RNA velocity analysis was used in this study[14]. Therefore, these functional gradients should only be interpreted as relative enrichment patterns and spatial organization within the latent state space. They should not be interpreted as temporal trajectories, state-transition paths, or lineage directionality.

2.8 Prognostic modeling using *RAC1*-responsive genes

To evaluate whether *RAC1*-responsive genes retained clinical relevance, they were mapped to the TCGA-LUAD bulk RNA-seq cohort for survival analysis[15]. TCGA-LUAD expression and clinical data were downloaded and processed using TCGAbiolinks[16]. Gene annotation, duplicated-gene removal, log2 transformation, and integration with overall survival time and survival status were performed before survival modeling.

Because not all *RAC1*-responsive genes were robustly detectable in the TCGA-LUAD cohort, candidate prognostic genes were further filtered. Genes were first selected from significantly differentially regulated DR genes with adjusted p value < 0.05, then restricted to genes detectable in the TCGA-LUAD expression matrix. After filtering low-expression genes, the top 20 genes ranked by perturbation distance were used as final candidate prognostic genes.

Univariate Cox regression was first performed on candidate genes, and genes with p value < 0.1 were passed to LASSO-Cox regression for feature selection. Non-zero coefficient genes at lambda.min were extracted to define the final prognostic signature[17]. A multivariate Cox model based on the selected genes was then used to calculate a risk score for each patient.

Patients were stratified into high-risk and low-risk groups according to the median risk score. Kaplan–Meier survival analysis was used to compare overall survival between the two groups. To evaluate whether the risk score had independent prognostic relevance, multivariate Cox regression was performed by integrating the risk score with clinical variables including age, sex, and pathological stage. Finally, time-dependent ROC curves and C-index were used to assess predictive discrimination.

3. Results

3.1 Virtual perturbation screening prioritizes *RAC1* as the major network-disruptive target

To identify the key perturbation target for downstream analysis, in silico virtual knockout screening was performed across multiple candidate genes. Among the successfully compared virtual perturbation targets, *RAC1* showed the strongest network-level response. Specifically, *RAC1*-vKO produced the highest mean distance (3.44 × 10^-5), the highest maximum distance (0.001713), and the largest number of significant DR genes (n = 13). In comparison, the mean distances for *AREG, LPCAT1, TMSB10, CTSD*, and *LDH*A were 3.20 × 10^-5, 2.38 × 10^-5, 2.23 × 10^-5, 1.63 × 10^-5, and 1.61 × 10^-5, respectively

These results indicate that *RAC1* virtual perturbation was associated with the strongest overall network response among the evaluated candidate genes. *RAC1* was therefore selected as the core perturbation target for subsequent GRN rewiring analysis, latent manifold reconstruction, and prognostic relevance analysis.

3.2 *RAC1* virtual perturbation reshapes regulatory network topology

After selecting *RAC1* as the core perturbation target, WT and *RAC1*-vKO GRNs were compared to evaluate how *RAC1* virtual perturbation reshaped regulatory network topology (Figure 1A). After thresholding and removing isolated nodes, the *RAC1*-vKO-induced GRN rewiring network retained 141 non-isolated genes and 140 rewired regulatory edges. Among these edges, 108 showed negative differential regulation and 32 showed positive differential regulation, suggesting that *RAC1* virtual perturbation was mainly associated with weakened regulatory connectivity, while a subset of connections became stronger.

Hub-status annotation showed that the rewiring network contained 13 new hub genes, 1 lost hub gene, and 127 unchanged nodes. *RAC1* itself was classified as the lost hub, with a delta degree of -34, indicating that its network connectivity substantially decreased after virtual knockout. Meanwhile, several genes displayed newly emerging hub-like connectivity in the *RAC1*-vKO network, suggesting that *RAC1* perturbation was associated not only with altered *RAC1* connectivity, but also with redistribution of network hub status among other genes.

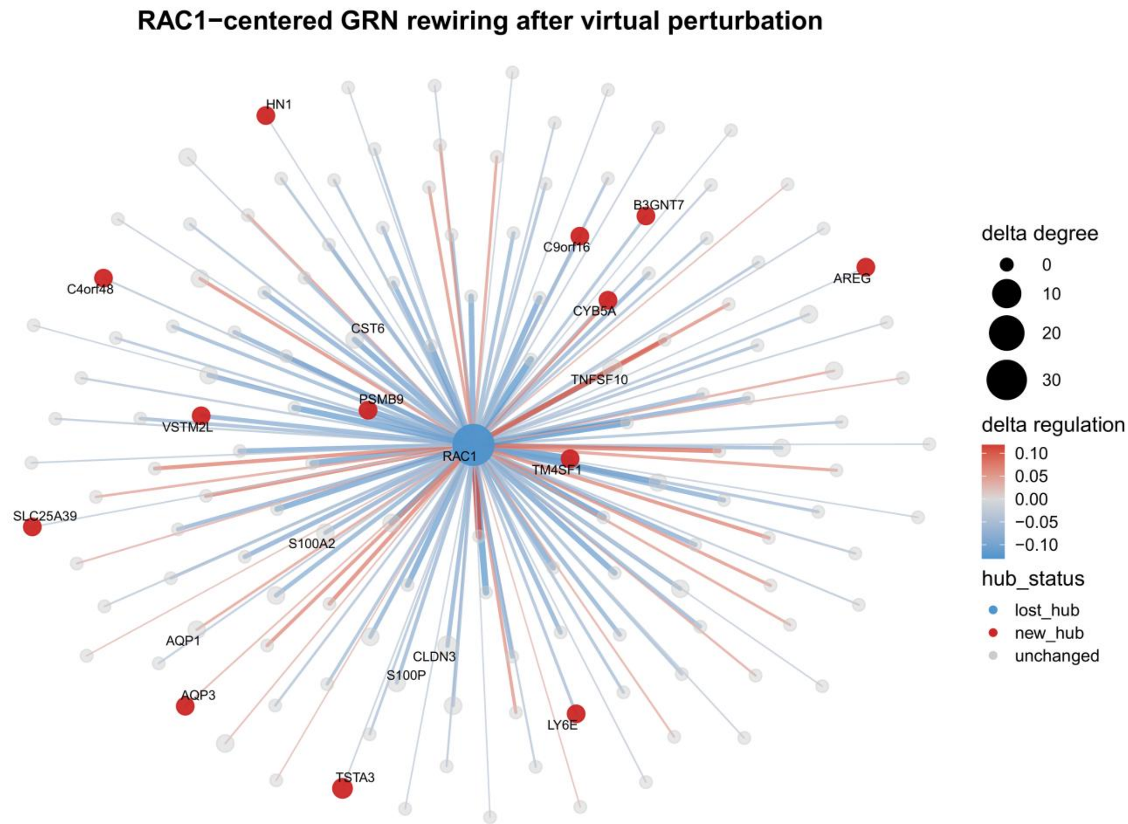


Figure 1A. *RAC1*-centered GRN rewiring after virtual perturbation. *RAC1*-vKO-induced GRN rewiring network constructed by comparing WT and *RAC1* virtual knockout regulatory networks. Nodes represent genes, and edges represent rewired regulatory interactions. Node size indicates absolute degree change, while node color indicates hub status, including new hub, lost hub, and unchanged nodes. Edge color represents the direction of differential regulation, with red indicating positive delta regulation and blue indicating negative delta regulation. Edge width reflects the absolute magnitude of differential regulation. This panel shows *RAC1*-vKO-associated regulatory topology rewiring and hub-status changes.

3.3 *RAC1*-responsive genes are embedded in a cross-program expression coordination network in WT tumor cells

After identifying *RAC1*-vKO-associated GRN rewiring, this project further examined whether *RAC1*-responsive genes already showed coordinated expression relationships in unperturbed WT tumor cells (Figure 1B). The expression coordination network constructed from top *RAC1*-responsive genes and related functional program genes contained 24 genes and 46 correlation-based edges.

Functional annotation of nodes showed that the network covered multiple programs, including alveolar identity, plasticity, epithelial, stress response, oncogenic activation, and remodeling programs. These results suggest that *RAC1*-responsive genes are not merely isolated marker-like genes, but are embedded within a cross-program expression coordination structure in WT tumor cells. This baseline coordination architecture provides a functional basis for the subsequent reconstruction of a latent tumor-state manifold.

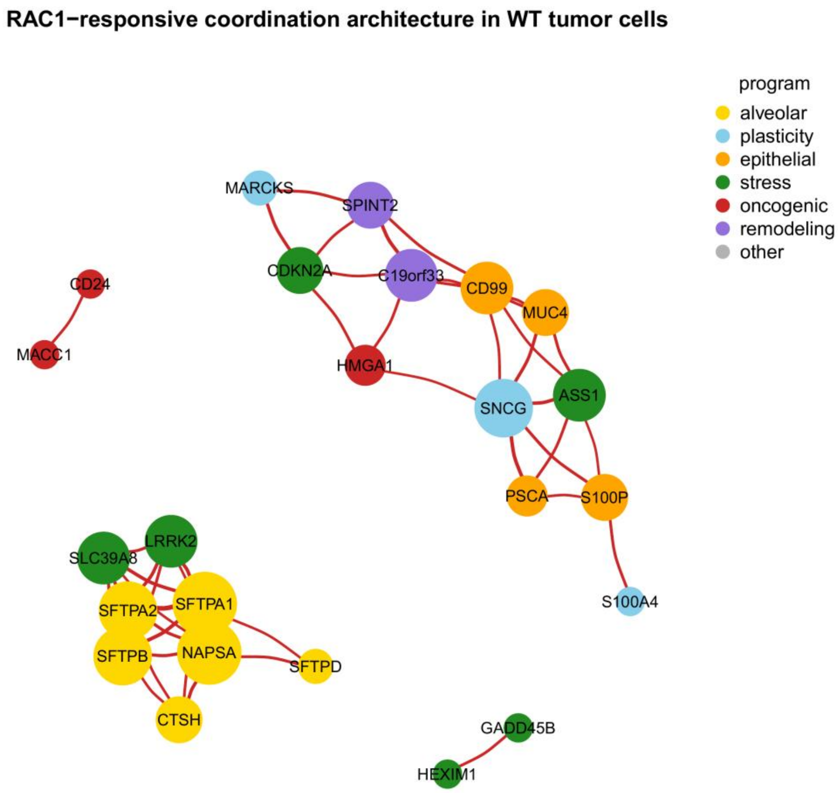


Figure 1B. Functional coordination network of *RAC1*-responsive genes in WT tumor cells. Correlation-based coordination network constructed using top *RAC1*-responsive genes, excluding *RAC1*, and predefined functional program genes in WT tumor cells. Nodes are colored by marker-based functional programs, and node size reflects network degree. Edge width represents correlation strength. This panel shows that *RAC1*-responsive genes are embedded in coordinated expression relationships across alveolar identity, plasticity, epithelial, stress response, oncogenic activation, and remodeling-related programs.

3.4 *RAC1*-responsive genes define a perturbation-informed latent tumor-state manifold

After characterizing *RAC1*-vKO-associated GRN rewiring and baseline expression coordination in WT tumor cells, we next examined whether *RAC1*-responsive genes could reveal structured tumor-state heterogeneity in unperturbed WT tumor cells. Based on top-ranked DR genes from *RAC1*-vKO analysis, 99 *RAC1*-responsive genes were retained after excluding *RAC1* itself and used to reconstruct a latent tumor-state manifold.

UMAP visualization showed that 5,000 WT tumor cells formed a structured latent state distribution within this *RAC1*-responsive feature space (Figure 2A). Based on cluster-level UCell functional program activities, 10 clusters were summarized into 7 hybrid functional states: alveolar-stress, alveolar-oncogenic, oncogenic-plasticity, oncogenic-remodeling, epithelial-plasticity, plasticity-remodeling, and epithelial-remodeling. Specifically, C0–C3 were annotated as alveolar-stress states; C4 as an alveolar-oncogenic state; C5 as an oncogenic-plasticity state; C6 as an oncogenic-remodeling state; C7 as an epithelial-plasticity state; C8 as a plasticity-remodeling state; and C9 as an epithelial-remodeling state.

These results suggest that *RAC1*-responsive genes captured structured tumor-cell heterogeneity within a perturbation-informed feature space, revealing a latent architecture composed of multiple hybrid functional states.

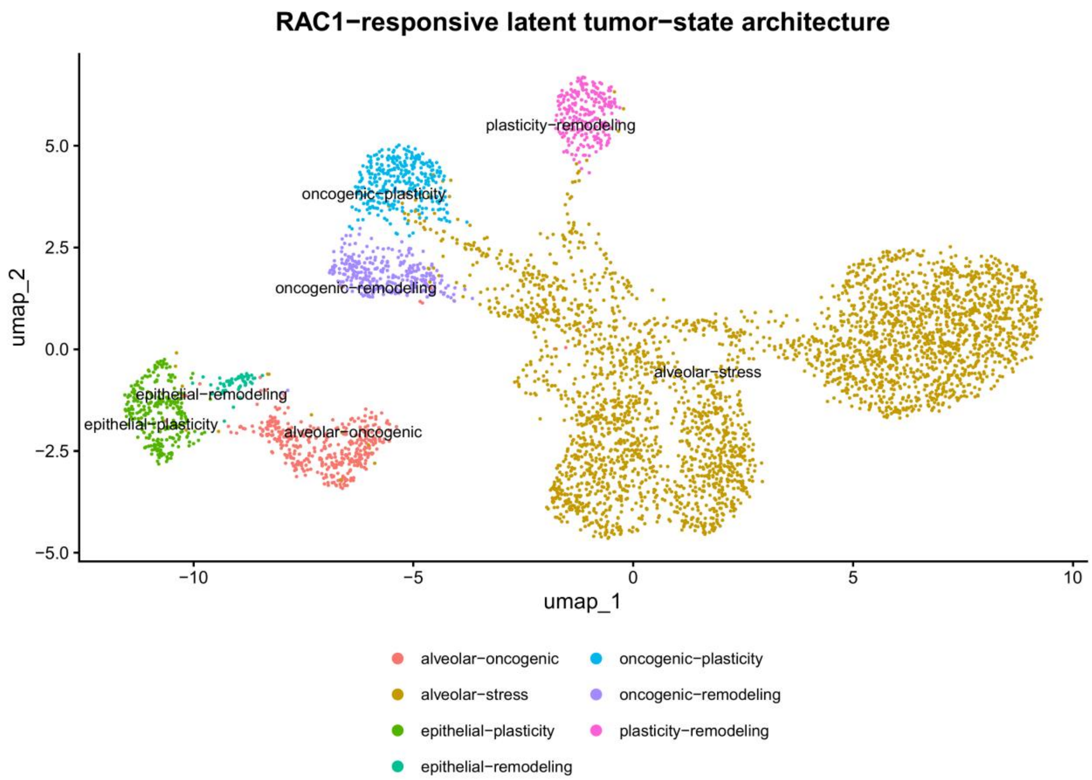


Figure 2A. Perturbation-informed latent tumor-state manifold. UMAP visualization of WT tumor cells reconstructed using 99 *RAC1*-responsive genes derived from *RAC1*-vKO-associated differential regulation analysis. Cells are colored by hybrid functional state annotations inferred from cluster-level UCell program activities. This panel shows that *RAC1*-responsive genes stratify WT tumor cells into multiple hybrid functional states within a perturbation-informed latent feature space.

3.5 Functional program scoring reveals hybrid tumor-state organization

To further interpret the functional meaning of different regions within the latent state manifold, UCell functional program scores were projected onto the same perturbation-informed state space (Figure 2B). Six functional programs were analyzed: alveolar identity, plasticity, epithelial, remodeling, stress adaptation, and oncogenic activation.

The results showed spatially organized distributions of these programs across the manifold. The alveolar identity program was relatively enriched in several alveolar-stress clusters, particularly C0 and C2, where mean UCell scores reached 0.912 and 0.965, respectively. The stress response program also remained relatively high in these regions, supporting the alveolar-stress hybrid state annotation.

In contrast, oncogenic activation, plasticity, and remodeling programs were relatively enriched in other regions of the manifold. For example, C5 showed higher oncogenic activation score (0.636) and plasticity score (0.520), consistent with an oncogenic-plasticity state. C6 showed high remodeling score (0.927) and oncogenic activation score (0.626), consistent with an oncogenic-remodeling state. C8 showed high remodeling score (0.801) and plasticity score (0.554), consistent with a plasticity-remodeling state.

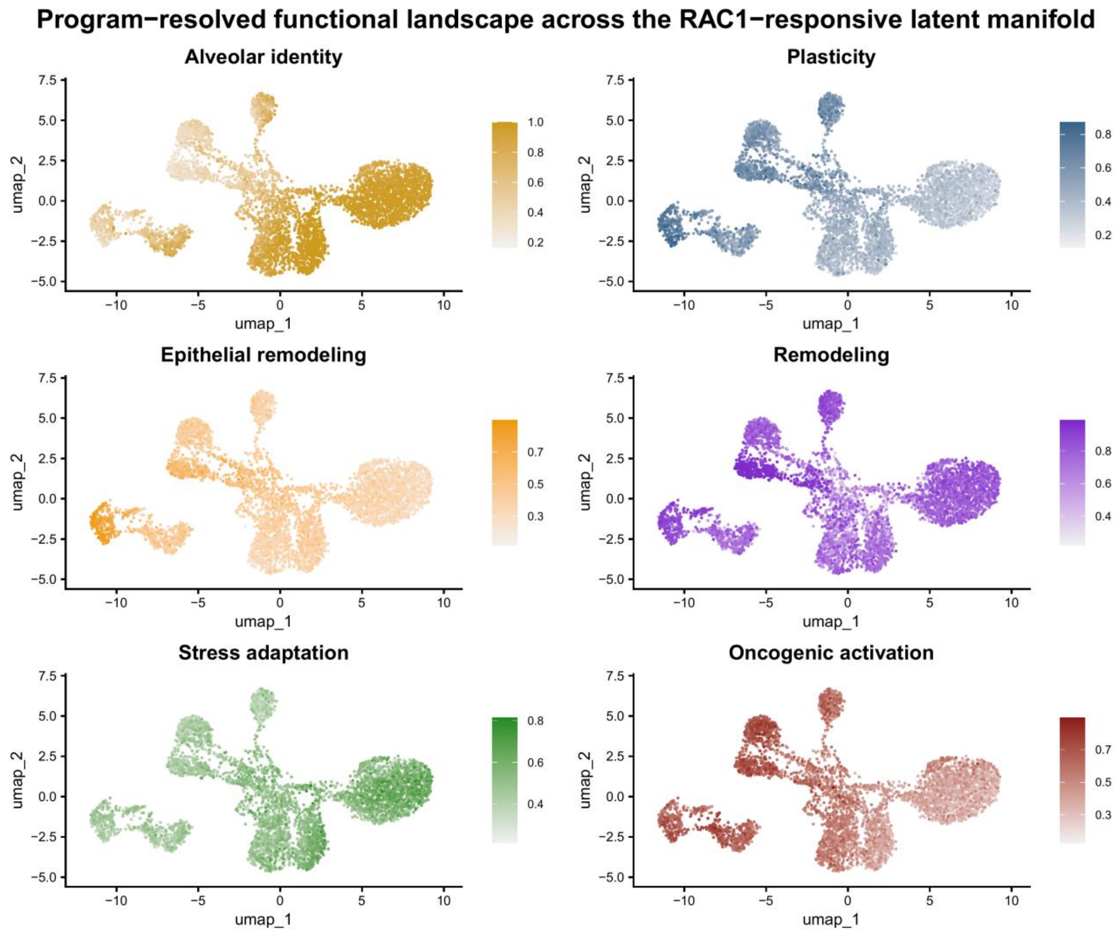


Figure 2B. Program-resolved functional landscape across the *RAC1*-responsive latent manifold. UCell-based functional program activities projected onto the perturbation-informed latent manifold. Individual panels show alveolar identity, plasticity, epithelial remodeling, remodeling, stress adaptation, and oncogenic activation. The spatial enrichment patterns show that *RAC1*-responsive states are associated with regionally organized functional programs within the *RAC1*-responsive feature space.

The cluster-program heatmap more clearly summarized the mean UCell program scores across clusters (Figure 2C). This heatmap further supported the hybrid state annotations shown in Figure 2A. Clusters annotated as alveolar-stress displayed relatively high alveolar identity and stress response scores, while C5, C6, and C8 showed different combinations of oncogenic activation, plasticity, and remodeling programs, corresponding to oncogenic-plasticity, oncogenic-remodeling, and plasticity-remodeling states, respectively.

Together, these results suggest that the *RAC1*-responsive latent state space is not defined by a single functional program or by fully discrete cell identities. Instead, it is organized by combinatorial functional program activity.

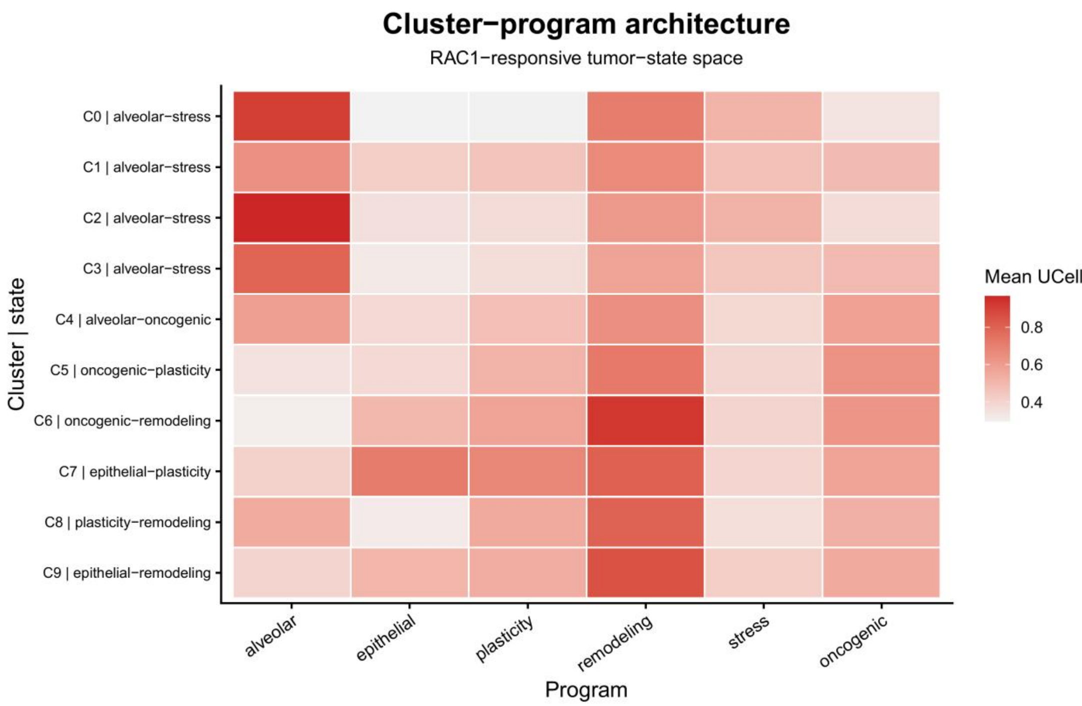


Figure 2C. Cluster-program architecture across *RAC1*-responsive latent tumor states. Heatmap showing mean UCell program scores across clusters in the perturbation-informed latent manifold. Rows represent cluster-state annotations, and columns represent functional programs. This panel summarizes how hybrid tumor-state annotations are associated with different combinations of alveolar, epithelial, plasticity, remodeling, stress, and oncogenic programs.

3.6 Functional gradients reveal continuous organization within *RAC1*-responsive tumor states

After knowing that different cell states were characterized by multiple functional programs, this project further examined whether these programs displayed continuous organization within the latent state space. Three functional program gradients were mapped onto the same manifold: plasticity–alveolar, stress–oncogenic, and remodeling–epithelial gradients (Figure 2D).

The plasticity–alveolar gradient showed clear spatial differences between plasticity-enriched regions and alveolar identity-enriched regions, with intermediate regions showing gradual changes in relative program enrichment. The stress–oncogenic gradient showed distinct relative enrichment biases across different manifold regions. In comparison, the remodeling–epithelial gradient showed that remodeling-associated activity was relatively dominant across much of the latent state space, while lower or near-neutral gradient values were restricted to smaller regions.

Together, these gradient patterns support the view that *RAC1*-responsive tumor states do not correspond to fully discrete cell categories. Instead, they display continuous spatial organization along multiple functional program axes. Importantly, these gradients only represent relative enrichment differences in latent state space. They do not imply temporal directionality, state-transition paths, or lineage progression.

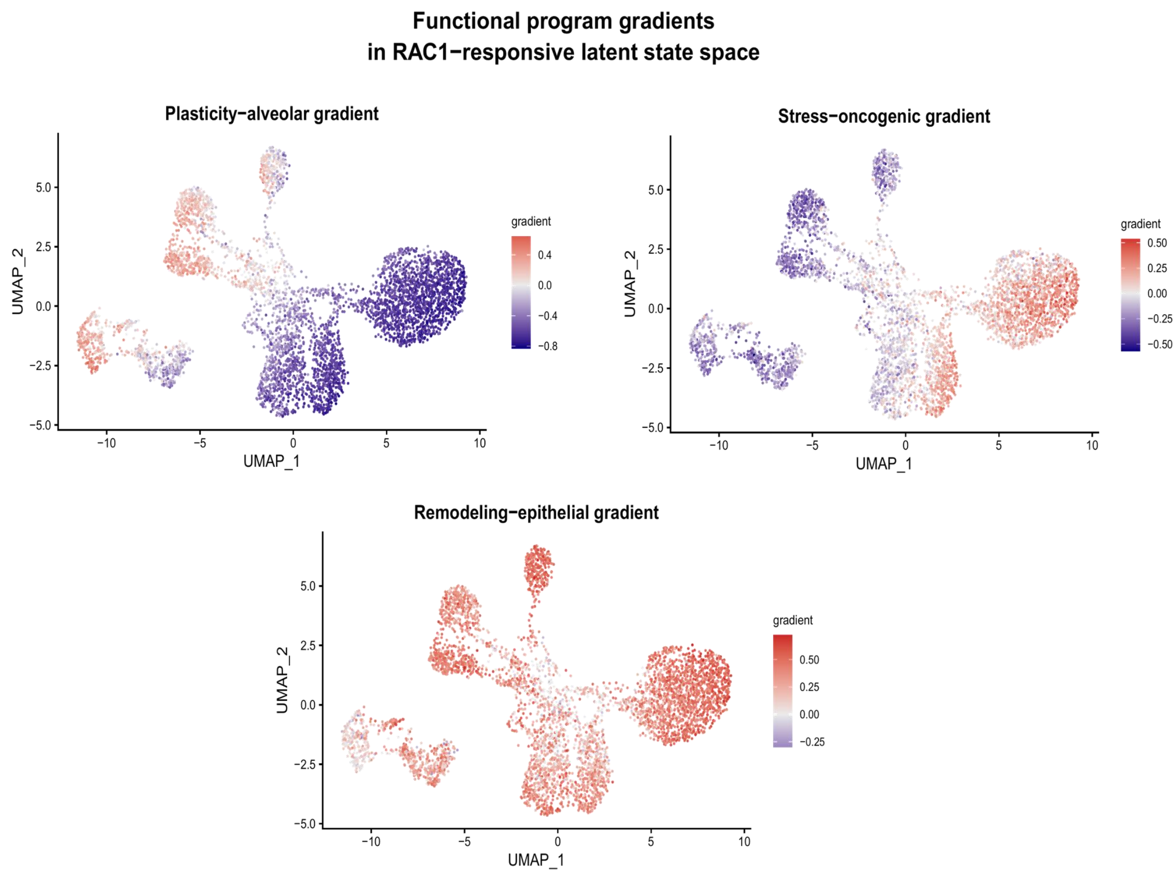


Figure 2D. Functional program gradients across *RAC1*-responsive latent state space. Signed functional gradients projected onto the *RAC1*-responsive latent manifold. Gradients were calculated by subtracting paired UCell program scores, including plasticity–alveolar, stress–oncogenic, and remodeling–epithelial gradients. These projections show continuous functional organization across the latent state space without implying temporal directionality, state-transition paths, or lineage progression.

3.7 *RAC1*-responsive genes retain prognostic relevance in TCGA-LUAD

After knowing that *RAC1*-responsive genes were associated with tumor-state organization at the single-cell level, this project further assessed whether these genes retained prognostic relevance in a clinical bulk RNA-seq cohort. *RAC1*-responsive genes were mapped to the TCGA-LUAD cohort, and prognostic modeling was performed after candidate gene filtering. LASSO-Cox regression identified an eight-gene *RAC1*-responsive prognostic signature consisting of *NAPSA, SFTPB, CTSH, CD99, CD24, PLAT, PSCA*, and *SFTPA1* (Figure 3A).

A multivariate Cox model based on these eight genes was used to calculate a risk score for each patient. Using the median risk score as the cutoff, 671 TCGA-LUAD patients were stratified into a high-risk group (n = 334) and a low-risk group (n = 337), with a median risk score of 0.969. Kaplan–Meier survival analysis showed that patients in the high-risk group had significantly worse overall survival than those in the low-risk group (Figure 3B), indicating that the *RAC1*-responsive prognostic signature could stratify LUAD patients by survival risk.

The risk score was further integrated with clinical variables including age, sex, and pathological stage to assess independent prognostic relevance. Multivariate Cox analysis showed that the risk score remained an independent prognostic factor after adjustment for clinical covariates (HR = 2.11, 95% CI = 1.74–2.57, p = 8.03 × 10^-14). Time-dependent ROC analysis further evaluated the discriminative ability of the signature at 1, 3, and 5 years. The model showed moderate discrimination for 1-year and 3-year survival prediction, while the 5-year AUC was relatively lower (Figure 3C). The C-index was 0.656 (95% CI: 0.614–0.699), indicating moderate overall discriminative ability.

Overall, these results suggest that *RAC1*-responsive genes are not only associated with LUAD tumor-state organization at the single-cell level, but also retain potential prognostic relevance in clinical bulk tumor cohorts.

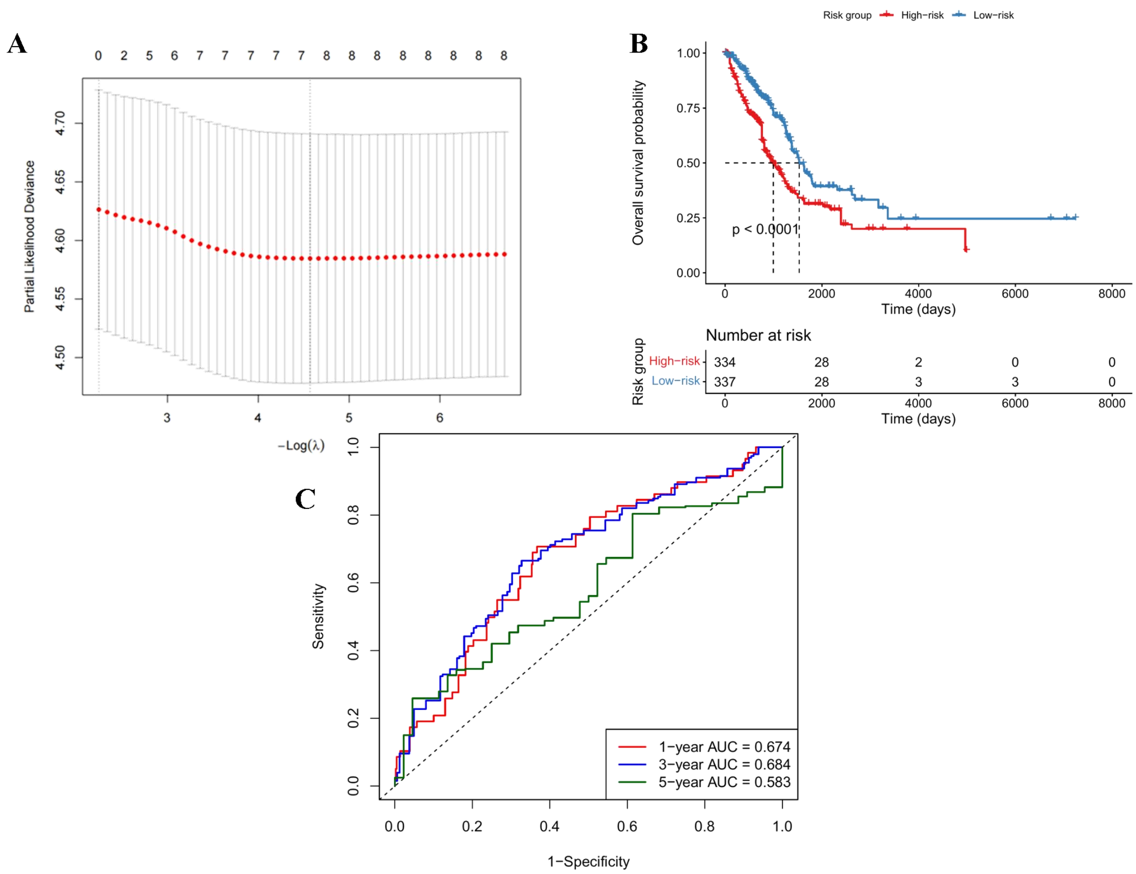
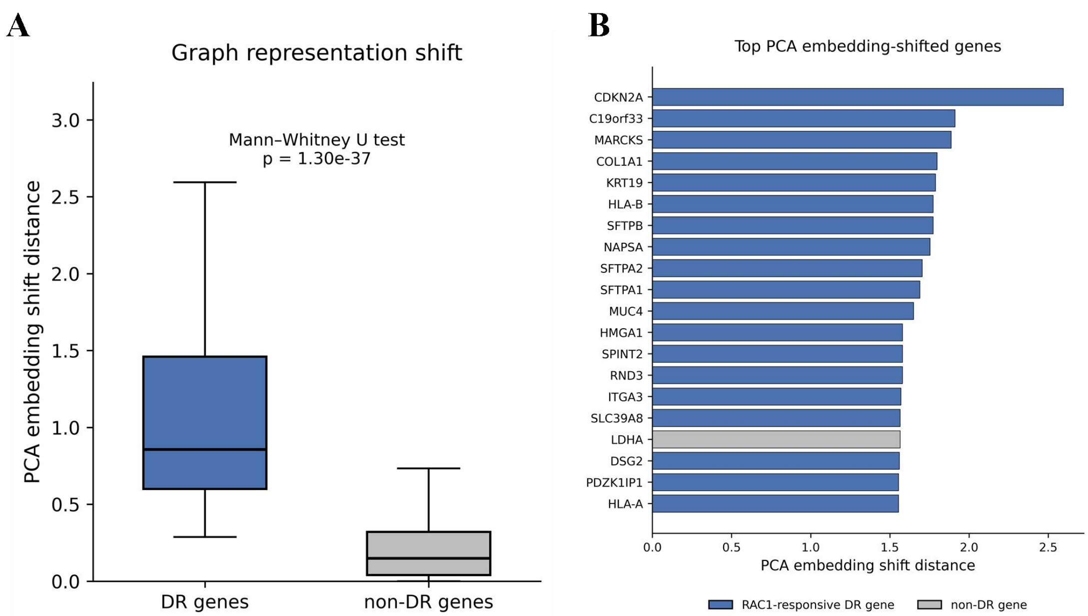


Figure 3. Prognostic relevance of *RAC1*-responsive genes in the TCGA-LUAD cohort. **(A) LASSO-Cox selection of *RAC1*-responsive prognostic genes.** LASSO-Cox regression analysis identifying prognostic *RAC1*-responsive genes in the TCGA-LUAD cohort. The final prognostic signature included *NAPSA, SFTPB, CTSH, CD99, CD24, PLAT, PSCA*, and *SFTPA1*, linking perturbation-responsive regulatory features to survival-associated gene programs. **(B) Kaplan–Meier survival analysis of *RAC1*-responsive risk groups.** Kaplan–Meier overall survival curves comparing high-risk (n = 334) and low-risk (n = 337) TCGA-LUAD patients stratified by the *RAC1*-responsive prognostic signature. The survival separation indicates that *RAC1*-responsive regulatory genes retain clinical outcome relevance in bulk tumor cohorts. **(C) Time-dependent ROC analysis of the *RAC1*-responsive prognostic model.** Time-dependent ROC curves evaluating the predictive performance of the *RAC1*-responsive prognostic model at 1, 3, and 5 years in the TCGA-LUAD cohort. AUC values are indicated in the figure legend. The model achieved a C-index of 0.656 (95% CI: 0.614–0.699), indicating moderate overall discriminative ability.

4. Supplementary Analysis: Python-based graph representation validation

As a supplementary analysis, the WT and *RAC1*-vKO regulatory networks generated by scTenifoldKnk were further examined using graph representation methods. This analysis was designed as a secondary computational validation of the existing WT/*RAC1*-vKO GRNs rather than an independent network inference approach. Across multiple graph-based strategies, *RAC1*-responsive DR genes consistently exhibited larger network-role shifts than non-DR genes. A composite topology shift score integrating degree centrality, weighted strength, betweenness centrality, and clustering coefficient was significantly higher in DR genes than in non-DR genes (Mann–Whitney U test, p = 1.10 × 10^-10). Similarly, when genes were represented by their network adjacency profiles and projected into a PCA-based graph representation space, DR genes showed substantially larger embedding shifts (p = 1.30 × 10^-37). Consistent results were also obtained using aligned node2vec embeddings, where DR genes again displayed significantly greater representation changes between WT and *RAC1*-vKO networks (p = 8.06 × 10^-10).

These findings are consistent with the scTenifoldKnk results and suggest that *RAC1*-responsive genes undergo broader changes in their regulatory positions following *RAC1* perturbation. Overall, this supplementary analysis provides additional graph-based support for *RAC1*-induced GRN rewiring and demonstrates the potential value of graph representation methods for studying regulatory network dynamics.



Supplementary Figure S1. Graph representation analysis supports network-role shifts of DR genes.
(A) PCA embedding shifts derived from adjacency-vector representations of WT and *RAC1*-vKO regulatory networks. DR genes exhibited significantly larger embedding shifts than non-DR genes (Mann–Whitney U test, p = 1.30 × 10^-37). **(B) Top genes ranked by PCA embedding shift.** Blue bars indicate DR genes and gray bars indicate non-DR genes. Genes with the largest embedding shifts were predominantly DR genes, consistent with stronger network-role changes between WT and *RAC1*-vKO networks..

5. Discussion

The central goal of this project was not to revalidate whether *RAC1* is associated with LUAD progression or prognosis. Instead, this project used *RAC1* as a virtual perturbation anchor to examine whether perturbation-associated regulatory network rewiring could be transformed into interpretable computational features for understanding LUAD tumor-state organization. In this framework, *RAC1*-vKO-derived DR genes were converted into perturbation-informed features, linking GRN rewiring, latent tumor-state manifold reconstruction, functional program organization, and prognostic relevance analysis within a unified computational workflow. *RAC1* was therefore not treated merely as a marker gene or prognostic gene, but as a computational entry point connecting virtual perturbation, network response, and single-cell state mapping.

Multi-gene virtual knockout screening prioritized *RAC1* as the candidate perturbation target with the strongest overall network response among the evaluated genes. *RAC1*-vKO-associated GRN rewiring further showed that *RAC1* virtual perturbation was associated not only with reduced *RAC1* network connectivity, but also with hub-status changes and rewired regulatory edges involving other genes. These findings suggest that *RAC1*-responsive network changes may be related to broader transcriptional coordination and regulatory network topology organization in LUAD tumor cells, rather than reflecting a single isolated pathway.

A key methodological feature of this study was the conversion of *RAC1*-responsive genes from a perturbation output into a feature space for single-cell state mapping. *RAC1*-vKO-derived DR genes were not used only as a list of affected genes; instead, they were used to reconstruct a latent tumor-state manifold in WT tumor cells. Therefore, the manifold constructed in this study was not intended to represent global transcriptomic heterogeneity in the same way as a conventional HVG-based UMAP. Rather, it represented a *RAC1*-responsive feature space carrying perturbation-associated network information. This step is the key conceptual distinction between this project and conventional single-cell clustering or differential expression analysis.

UCell-based functional program scoring showed that clusters within the *RAC1*-responsive latent manifold were not explained by single functional programs alone. Instead, they displayed hybrid functional states such as alveolar-stress, oncogenic-plasticity, oncogenic-remodeling, and plasticity-remodeling states. Functional gradient analysis further suggested that these hybrid states were embedded within continuous spatial variation of program enrichment across the manifold. This supports the interpretation that *RAC1*-responsive tumor states are better understood as a continuous and combinatorial functional landscape rather than as fully discrete cell categories. Importantly, these gradients should not be interpreted as temporal trajectories, because this study did not use time-series data, lineage tracing, RNA velocity, or real *RAC1* perturbation single-cell datasets.

The TCGA-LUAD prognostic relevance analysis showed that an eight-gene signature derived from *RAC1*-responsive genes could stratify patients by survival risk and remained independently associated with overall survival after adjustment for clinical variables. This suggests that *RAC1*-responsive features were not only associated with tumor-state organization at the single-cell level, but also retained prognostic relevance in a bulk clinical cohort. However, this analysis was intended to evaluate the clinical relevance of perturbation-responsive features, rather than to establish a mature clinical prediction model.

In addition, a supplementary graph representation analysis provided further support for the network-level effects of *RAC1* perturbation. *RAC1*-responsive genes showed larger shifts in network topology and embedding space than non-responsive genes, suggesting more substantial changes in their regulatory roles following *RAC1* knockout. Although this analysis was based on the same inferred GRNs and should therefore be viewed as complementary rather than independent validation, it strengthens the overall evidence for *RAC1*-induced network rewiring and highlights the potential of graph-based approaches for studying gene regulatory systems.

Several limitations should be noted. First, *RAC1* perturbation in this study was based on computational virtual knockout rather than experimental CRISPR knockout, RNA interference, or other real perturbation data. Therefore, the biological conclusions require further validation using experimental perturbation systems. Second, this study did not incorporate time-series data, lineage tracing, RNA velocity, or spatial transcriptomics, and therefore cannot infer true cell-fate direction or spatiotemporal dynamics. Third, the functional programs used here were manually curated from known marker genes, and UCell scoring was performed within the *RAC1*-responsive feature space. Thus, the resulting program scores are best interpreted as relative functional tendencies among *RAC1*-responsive genes, rather than complete pathway activities at the whole-transcriptome level.

Overall, this project demonstrates a computational biology workflow that starts from virtual perturbation and connects regulatory network rewiring, single-cell latent state mapping, and clinical relevance projection.

6. Conclusion

This study developed a perturbation-informed computational framework centered on *RAC1* virtual perturbation in LUAD tumor cells. *RAC1*-vKO-associated GRN rewiring information was converted into perturbation-responsive features and used to reconstruct a latent tumor-state manifold in unperturbed tumor cells. This manifold revealed hybrid functional states and continuous functional gradients, while *RAC1*-responsive genes also retained prognostic relevance in the TCGA-LUAD cohort. Also, Supplementary analysis further supported stronger network-role shifts of *RAC1*-responsive genes between WT and RAC1-vKO regulatory networks.

Rather than focusing on *RAC1* expression or prognostic modeling alone, this project illustrates how virtual perturbation and regulatory network reconstruction can be integrated with single-cell state mapping. These findings suggest that *RAC1*-responsive regulatory rewiring may be associated with transcriptional coordination and tumor-state organization in LUAD, providing a systems-level computational strategy for studying tumor-cell plasticity from the perspective of network perturbation.

Data and Code Availability

All analysis scripts, processed intermediate result tables, and manuscript figures of this project are deposited in the GitHub :https://github.com/likejin6/RAC1-latent-state-mapping

Single-cell RNA-seq data were acquired from the GEO dataset GSE131907 https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE131907；

Transcriptomic and clinical data of TCGA lung adenocarcinoma (LUAD) were downloaded using the TCGAbiolinks R package.

Large raw data files and heavy intermediate R objects are not included in the repository due to file-size limitations, but the analysis scripts document how each processed file and figure was generated.

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